TEXORA · LECTURE SHEET

Preparation of Print Paste and General Flow Chart of the Printing Process

The paste is a chemical system, not coloured glue: thickener choice, component order, and the eight-step flow from prepared fabric to inspected print.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 8 · LEVEL 3 TERM 1

1. Printing versus dyeing — localised coloration

Printing applies the same colorants as dyeing, but only where the design demands, from a viscous **print paste** that stays where the screen or roller puts it until fixation drives the dye into the fibre. Paste rheology is therefore half the technology: too thin and the design bleeds (flushing); too thick and it will not transfer or penetrate. Print pastes are shear-thinning — they flow under the squeegee and recover viscosity at rest.

2. Anatomy of a print paste

COMPONENT	FUNCTION AND TYPICAL CHOICE
Dye or pigment	Colour: reactive, vat, disperse or pigment systems per fibre (Topics 9–10)
Thickener	Gives print definition: sodium alginate for reactives (it does not react with the dye), starch/guar and synthetic polyacrylates for pigments, emulsion thickeners historically for pigment (kerosene-based, now replaced for VOC reasons)
Water	Solvent and thickener medium; the paste is typically 50–80% water before thickener stock is counted
Hygroscopic / solvent assistant	Urea (50–150 g/kg) dissolves dye and holds moisture during steaming; glycerine in small amounts
Fixation chemicals	Alkali (sodium bicarbonate for reactives), reducing agents for vat prints (sodium formaldehyde sulphoxylate), acid donors for disperse/nylon
Auxiliaries	Defoamer, wetting agent, resist/oxidant protectors (e.g. Ludigol for reactive prints), softener for pigment pastes

A standard reactive print paste (per kg): thickener stock 500 g + urea 100 g + $NaHCO_3$ 25 g + resist salt 10 g + dye X g + water to 1,000 g

The thickener choice is chemistry, not habit: alginate's carboxyl groups are largely unionised/inert towards reactive dyes, whereas starch and CMC carry hydroxyls that compete with the fabric for the dye — paste wasted on thickener is fixation lost.

3. Making the paste — order and discipline

1. Prepare thickener stock first (alginate sprinkled into agitated water, rested to hydrate fully — lumps print as specks).
2. Dissolve dye separately in hot water with urea; strain through fine mesh.
3. Blend dye solution into thickener stock; add alkali last (for reactives, bicarbonate is mild enough for paste storage; stronger alkali would hydrolyse dye in the pot).
4. Rest, de-aerate, check viscosity and pH; label with recipe, date and shade. Paste life: reactive pastes 2–5 days cool-stored; pigment pastes weeks.

4. General flow chart of the printing process

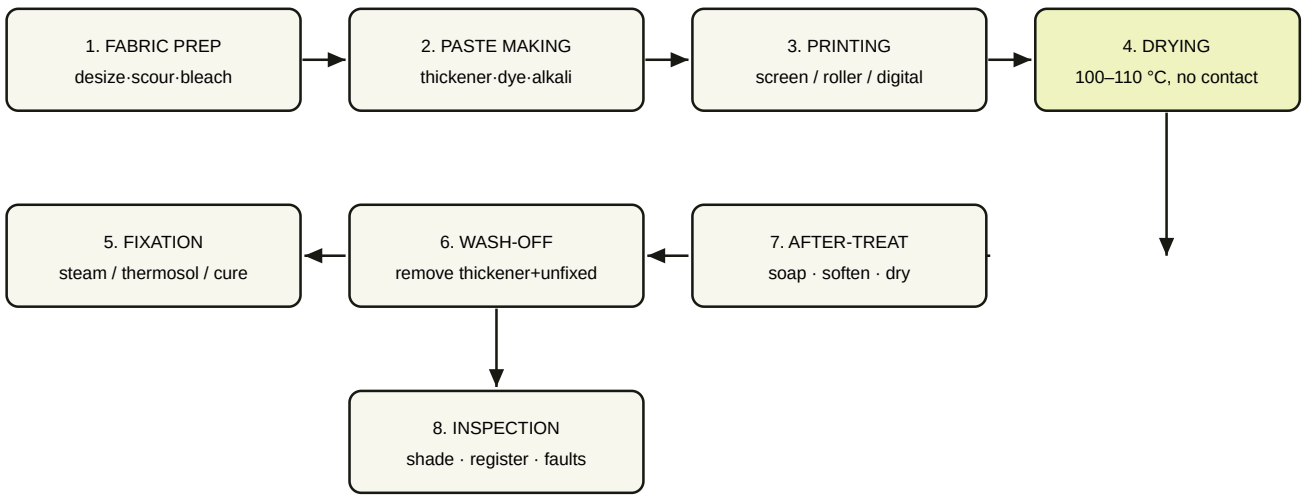


Fig. 1 — General flow chart of textile printing, from prepared fabric to inspected print.

Reading the chart right to left exposes the logic: inspection defines acceptance; wash-off only succeeds if fixation was complete; fixation only works on a properly dried print; the print is only as sharp as the paste. **Fixation routes:** saturated steam 100–105 °C, 5–10 min (reactives on cotton, with urea holding moisture for dye diffusion); superheated steam/thermosol 180–200 °C (disperse on polyester); dry-heat cure 140–160 °C (pigments — binder crosslinking, Topic 10).

5. Wash-off — the printer's soaping

After fixation the fabric carries thickener film, unfixed/hydrolysed dye and chemicals. Wash-off (cold rinse → warm → hot soaping → rinse) must remove all of it without letting loose dye stain white grounds — anti-redeposition agents and progressive bath heating protect the ground. A print that passes dry inspection but fails wash-off has failed: the flow chart's step 6 is a quality gate equal to printing itself.

6. Glossary — key terms

- **Print paste:** viscous, shear-thinning vehicle carrying colorant and fixation chemicals.
- **Flushing:** spreading of the print beyond the design edge from low viscosity or over-wet fabric.
- **Register:** exact alignment of successive colours of a design.
- **Ground:** the unprinted area, whose whiteness wash-off must protect.

7. Key points

- Thickener is a chemical choice: alginate for reactives because it does not compete for the dye.
- Alkali goes into reactive paste last, as mild bicarbonate, to protect paste life.
- Print → dry → fix → wash-off: no step can compensate for the previous one.

8. Summary

Print paste formulation and the eight-step flow chart together define printing as localised dyeing with a rheology problem attached. Every later printing topic — methods, styles, pigment systems — is a variation on this skeleton.

9. Practice

1. Make up 2 kg of the standard reactive paste in Section 2 with dye at 40 g/kg: list every component mass.
2. Why is alginate preferred over starch for reactive printing? Answer in terms of fixation loss.
3. A print is sharp off the machine but the white ground stains in wash-off. Give two causes and two remedies.

Answer key

1. Per kg: stock 500 g, urea 100 g, $NaHCO_3$ 25 g, resist salt 10 g, dye 40 g, water 1000−675 = 325 g. For 2 kg, double all: 1000, 200, 50, 20, 80, 650 g (total 2,000 g ✓).

2. Starch/CMC hydroxyls react with reactive dye exactly as cellulose does, consuming dye that can then never reach the fabric — and washing out as coloured thickener. Alginate's carboxylate structure is essentially unreactive to the dye, so fixation goes to the cloth.
3. Causes: incomplete fixation (too much genuinely unfixed dye) and thickener redeposition carrying dye onto the ground. Remedies: verify steam time/urea level for full fixation; use anti-redeposition agent, start wash-off cold and dilute, and soap thoroughly before any hot bath sets the stain.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib